

Parent Guide — primary / module-14-magic-squares

IKS Curriculum

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Parent Guide — Module 14: Magic Squares (Primary)

Your child's age: Primary (Grades 3–5) · **Module length:** 10 sessions × ~30 min

What your child is learning

The puzzle: put the numbers 1 to 9 in a 3×3 grid so every row, every column, and both diagonals all add up to the same number. That number turns out to be 15. We call this a **magic square**.

Why “magic”: because every line in the square adds up to the same total — and that feels surprising. (It is not magic. It is careful arrangement.)

Stories your child will hear: - The **Lo Shu turtle** — a Chinese story about an emperor who saw a 3×3 numbered pattern on a turtle's shell coming out of a river. - The **Khajuraho stone** — a real carving on a temple wall in central India, showing a 4×4 magic square.

We treat the stories as stories. The carvings are real artefacts. The numbers are numbers.

What to expect this fortnight

Daily chant: “*All to fifteen, all to fifteen!*” — your child will sing this at home. It helps the addition stick.

Tile play: your child uses paper tiles or tokens labelled 1–9 to try arrangements. Most days at school there is at least one hands-on session.

Make-and-Keep (Day 9): each child decorates their own 3×3 magic square page. This comes home on Day 10. Praise the effort, not just the result.

How to help

Helpful: - Ask: “Show me a row of your magic square that adds to 15.” Let your child do the addition. - Play “make 10”: call out a single digit; your child says the partner that makes 10 (e.g. you say “7,” they say “3”). 2 minutes a day is plenty. - Read or listen with your child to the Lo Shu turtle story.

Not helpful: - Showing your child the “answer” online. The discovery is the lesson. - Pushing for 4×4 or bigger. That’s Middle-band territory. - Treating the magic square as magical or spiritual at this age. We keep it mathematical.

What success looks like

By the end of the module, your child can: - Add three single-digit numbers and check whether the sum is 15. - Arrange the numbers 1 to 9 in a 3×3 to make a magic square (with help on early tries; independently by Day 9 for most children). - Tell the Lo Shu turtle story in 3–4 sentences. - Point to Madhya Pradesh on a map of India and say “the Khajuraho stone is there.” - Sing the “all to fifteen” chant.

A note on the cultural framing

Magic squares have been used in some traditional contexts as protective or auspicious symbols. In this curriculum we engage them as mathematics. If your child mentions hearing about them in a religious context at home, that’s wonderful — we simply approach them differently in the math class.

Want to go deeper?

Middle band (Grades 6–8) introduces the magic constant formula and Nārāyaṇa Paṇḍita’s stair-step method for building bigger magic squares. Preview: [curriculum/middle/module-14-magic-squares/README.md](#).

10-Day Lesson Plan

Module 14 (Primary) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / artefact
1	The Magic Number 15	days/day-01-magic-number.md	Wall chart “to 15”
2	Pairs that add to 10	days/day-02-pairs-to-ten.md	Tile pair game
3	The First Try	days/day-03-first-try.md	Worksheet: blank 3×3
4	The Secret of the Centre	days/day-04-secret-centre.md	Class discovery
5	Our Class Magic Square	days/day-05-class-square.md	Big chart on wall
6	Magic Square Sing-along	days/day-06-singalong.md	activities/activity-01-tile-game.md

Day	Session title	Lesson file	Activity / artefact
7	The Lo Shu Turtle	days/day-07-lo-shu-turtle.md	Story + draw a turtle
8	The Khajuraho Stone	days/day-08-khajuraho-stone.md	India map find
9	Make-and-Keep	days/day-09-make-and-keep.md	Decorate own square
10	Show and Tell	days/day-10-show-and-tell.md	Sticker assessment

Pacing notes

- Sessions are ~30 minutes. Be flexible — Days 3 and 4 can each stretch if children are still trying combinations.
- The chant (“all to fifteen, all to fifteen”) is sung at the start of every class from Day 2 onwards. Repetition is the lesson.
- Days 1–5 are the math; Days 6–8 add story and culture; Days 9–10 are the project.

Daily chant

“One, two, three — count to fifteen! Four, five, six — still fifteen! Seven, eight, nine — all to fifteen!”

(Use during settle-in. Clap on each number.)

What children leave with

- A coloured 3×3 magic square they made themselves.
- The chant “all to fifteen.”
- Two stories: the **Lo Shu turtle** and the **Khajuraho stone**.
- Confidence adding three single-digit numbers.

Homework

Homework — Magic Squares — Bhadrakanita (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — Discover the 3×3** · Draw about today’s concept. (~5 min)
2. **Day 2 — The Magic Constant** · Draw about today’s concept. (~5 min)
3. **Day 3 — How Many 3×3 Magic Squares?** · Draw about today’s concept. (~5 min)
4. **Day 4 — Stair-Step Method (Nārāyaṇa)** · Draw about today’s concept. (~5 min)
5. **Day 5 — Practice — 5×5 and 7×7** · Draw about today’s concept. (~5 min)

6. **Day 6 — 4×4 and the Khajuraho Square** · Draw about today's concept. (~5 min)
7. **Day 7 — Historical Context — Nārāyaṇa and Predecessors** · Draw about today's concept. (~5 min)
8. **Day 8 — Cross-Cultural — Lo Shu and Dürer** · Draw about today's concept. (~5 min)
9. **Day 9 — Design Your Own** · Draw about today's concept. (~5 min)
10. **Day 10 — Final Share and Assessment** · Draw about today's concept. (~5 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.